



IoT Software Analyst

QP Code: SSC/Q8205

Version: 4.0

NSQF Level: 5.5

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Qualification Pack

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SSC/Q8205: IoT Software Analyst

Brief Job Description

The IoT Software Analyst role, based on the modules you've posted, involves designing, developing, testing, and maintaining end-to-end IoT solutions. Key activities include working with connectivity protocols, developing and integrating software at device, platform, and cloud levels, ensuring system reliability, and testing for functionality and performance. The role also emphasizes automation (CI/CD pipelines), configuration management, quality assurance through various testing methods (functional, non-functional, end-to-end), and proper documentation. Skills required include programming, working with IoT platforms, managing source code, employing DevOps tools, and ensuring system security and compliance. Overall, the analyst is crucial in enabling seamless, reliable, and secure IoT systems from development to deployment.

Personal Attributes

IoT Software Analysts must be able to communicate effectively, build relationships with others and collaborate with various stakeholders. They must have strong analytical abilities and a zeal to learn new things.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8218: Design, develop and maintain software across appropriate layers of IoT solutions](#)
2. [SSC/N8219: Develop software applications and manage application lifecycle for IoT solutions](#)
3. [SSC/N8245: Develop tools, processes and mechanisms for continuous integration and delivery for IoT solutions](#)
4. [SSC/N8224: Develop simulations and manage tests for end-to-end IoT solutions](#)
5. [SSC/N8244: Understand functional & non-functional testing of IoT solutions](#)
6. [SSC/N8238: Create technical documents and manuals](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills

Qualification Pack

Occupation	Internet of Things
Country	India
NSQF Level	5.5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512
Minimum Educational Qualification & Experience	Completed 3 year UG degree (of 3Y/4Y UG.PG or UG or diploma with courses related to Engg./ Science) OR Completed 3 year diploma after 10th (PG or UG or diploma with courses related to Engg./ Science) with 3 Years of experience Relevant Field- Computer Science/ IT/IoT/Software Development The relevant experience would include work, internship and apprenticeship after completing relevant educational qualifications. OR Previous relevant Qualification of NSQF Level (5) with 1.5 years of experience Relevant Field- Computer Science/ IT/IoT/Software Development The relevant experience would include work, internship and apprenticeship after completing relevant educational qualifications.
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-5.5-IT-045972025-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8218: Design, develop and maintain software across appropriate layers of IoT solutions

Description

This unit is about designing, developing and maintaining software across the device, connectivity, platform, server, analytics, application, and presentation layers of the IoT solution.

Scope

The scope covers the following :

- Develop device layer
- Develop connectivity layer
- Develop platform layer
- Develop server & application layer
- Deploy streaming analytics
- Develop presentation layer
- Drive continuous monitoring

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** design, develop and maintain real-time operating systems at the device layer
- PC2.** design, develop and maintain device software which enables access to device hardware
- PC3.** design, develop and maintain device software which enables features such as remote controlling the device, upgrading its firmware, monitoring its battery level, etc.
- PC4.** design, develop and maintain firmware that enables wired/wireless connectivity between the end device and IoT gateway
- PC5.** apply appropriate wired/wireless connectivity protocols for device-device or device-gateway communications
- PC6.** design, develop and maintain IoT device management with Over the Air functionality
- PC7.** design, develop and maintain the deployment of analytics packet Over the Air updates on the devices through the IoT device management platform
- PC8.** evaluate IOT Device Management Commercial off-the-shelf platforms
- PC9.** design, develop and maintain applications to run on standard IoT platforms
- PC10.** design, develop and maintain software which enables IoT platforms to interact with large numbers of devices and gateways
- PC11.** design, develop and maintain a central registry to identify the devices/gateways running in an IoT solution
- PC12.** design, develop and maintain scalable processing solutions with the ability to store and analyze data
- PC13.** design, develop and maintain software applications, libraries, server software and APIs
- PC14.** Design and develop analytics solutions for the edge / platform layers

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- PC15.** develop & deploy a variety of streaming analytics / complex events / monitoring requirement for various use-cases
- PC16.** design, develop and maintain user experience interfaces across layers of the IoT solution
- PC17.** design, develop and maintain reports, graphs and visualization dashboards of IoT solution
- PC18.** continually evaluate software reliability and performance
- PC19.** use DevOps tools for automated deployment and monitoring of software applications
- PC20.** continually evaluate software reliability, power and performance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to developing and deploying software applications
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting the features of developed applications
- KU4.** who to involve while developing, maintaining, deploying, evaluating, and monitoring software
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** the range of code generation tools and unit testing tools used to develop software code
- KU7.** how to use coding tools
- KU8.** how to create, review and execute unit test cases
- KU9.** how to determine whether components are suitable for re-use
- KU10.** various types of problems and defects that may occur and how to resolve them
- KU11.** how to build a repository of corrective actions for each problems and defects to improve future designs
- KU12.** how to use corrective actions to optimize the performance of software code
- KU13.** how to identify and refer anomalies
- KU14.** how to work on various operating systems such as Linux, Ubuntu, or Windows
- KU15.** different data flows and structures
- KU16.** various algorithmic models and their design

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making process
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor

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- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** provide opinions on work in a detailed and constructive way
- GS10.** apply balanced judgments to different situations
- GS11.** check the work is complete and free from errors
- GS12.** work independently and collaboratively
- GS13.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. design, develop and maintain real-time operating systems at the device layer	3	4	-	1
PC2. design, develop and maintain device software which enables access to device hardware	1	2	-	1
PC3. design, develop and maintain device software which enables features such as remote controlling the device, upgrading its firmware, monitoring its battery level, etc.	2	3	-	1
PC4. design, develop and maintain firmware that enables wired/wireless connectivity between the end device and IoT gateway	2	3	-	1
PC5. apply appropriate wired/wireless connectivity protocols for device-device or device-gateway communications	2	3	-	1
PC6. design, develop and maintain IoT device management with Over the Air functionality	2	2	-	1
PC7. design, develop and maintain the deployment of analytics packet Over the Air updates on the devices through the IoT device management platform	1	2	-	1
PC8. evaluate IOT Device Management Commercial off-the-shelf platforms	1	2	-	1
PC9. design, develop and maintain applications to run on standard IoT platforms	2	4	-	1
PC10. design, develop and maintain software which enables IoT platforms to interact with large numbers of devices and gateways	1	2	-	1
PC11. design, develop and maintain a central registry to identify the devices/gateways running in an IoT solution	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. design, develop and maintain scalable processing solutions with the ability to store and analyze data	1	2	-	1
PC13. design, develop and maintain software applications, libraries, server software and APIs	2	4	-	1
PC14. Design and develop analytics solutions for the edge / platform layers	1	2	-	1
PC15. develop & deploy a variety of streaming analytics / complex events / monitoring requirement for various use-cases	1	2	-	1
PC16. design, develop and maintain user experience interfaces across layers of the IoT solution	1	2	-	1
PC17. design, develop and maintain reports, graphs and visualization dashboards of IoT solution	1	1	-	1
PC18. continually evaluate software reliability and performance	1	2	-	1
PC19. use DevOps tools for automated deployment and monitoring of software applications	1	2	-	1
PC20. continually evaluate software reliability, power and performance	3	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8218
NOS Name	Design, develop and maintain software across appropriate layers of IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	5.5
Credits	4
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8219: Develop software applications and manage application lifecycle for IoT solutions

Description

This unit is about identifying the requirements for the software application and developing, deploying, managing, and maintaining it.

Scope

The scope covers the following :

- Specify requirements
- Develop application
- Deploy applications
- Use Application Management Services
- Drive application maintenance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** gather requirements for the development of software applications from relevant stakeholders
- PC2.** ensure regulatory and compliance requirements are also identified
- PC3.** categorize specifications and prioritize them based on when they must be developed
- PC4.** design and develop software applications for the IoT solution
- PC5.** continuously verify that the application is in line with the specified requirements
- PC6.** identify appropriate methodologies (such as agile, scrum, etc.) to structure, plan, and control the solution development process
- PC7.** develop and run test cases on the application being built
- PC8.** plan the deployment model of the application
- PC9.** roll out the application when it is ready for release
- PC10.** use Application Management Services to provide ongoing application support to external providers
- PC11.** use Application Monitoring Services to delegate the responsibilities of monitoring and releasing patches, bug-fixes, and enhancements to external vendors
- PC12.** monitor and manage deployed application
- PC13.** continuously look for and resolve bugs in software application
- PC14.** plan for application updates and patches
- PC15.** plan for application upgrades
- PC16.** plan for the retirement of application and migration to a new application

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to developing, deploying, and managing software applications
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting the deployment procedures and the features of developed applications
- KU4.** who to involve while developing, managing, deploying, evaluating, and monitoring software
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** the range of code generation tools and unit testing tools used to develop software code
- KU7.** various development methodologies (such as agile, scrum, etc.)
- KU8.** application management services
- KU9.** application monitoring services
- KU10.** how to use coding tools
- KU11.** how to create, review and execute unit test cases
- KU12.** how to determine whether components are suitable for re-use
- KU13.** different types of problems and defects that may occur and how to resolve them
- KU14.** how to build a repository of corrective actions for each problem and defect to improve future designs
- KU15.** how to use corrective actions to optimize the performance of software code
- KU16.** how to identify and refer anomalies
- KU17.** how to work on various operating systems such as Linux, Ubuntu, or Windows
- KU18.** different data flows and structures
- KU19.** different algorithmic models and their design

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making process
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** ask for clarification and advice from appropriate people
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** provide opinions on work in a detailed and constructive way
- GS10.** apply balanced judgments to different situations
- GS11.** check the work is complete and free from errors
- GS12.** work independently and collaboratively
- GS13.** contribute to the quality of teamwork

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. gather requirements for the development of software applications from relevant stakeholders	4	3	-	2
PC2. ensure regulatory and compliance requirements are also identified	2	2	-	1
PC3. categorize specifications and prioritize them based on when they must be developed	1	3	-	1
PC4. design and develop software applications for the IoT solution	3	5	-	2
PC5. continuously verify that the application is in line with the specified requirements	3	4	-	1
PC6. identify appropriate methodologies (such as agile, scrum, etc.) to structure, plan, and control the solution development process	3	4	-	2
PC7. develop and run test cases on the application being built	1	2	-	1
PC8. plan the deployment model of the application	1	2	-	1
PC9. roll out the application when it is ready for release	1	2	-	1
PC10. use Application Management Services to provide ongoing application support to external providers	1	2	-	1
PC11. use Application Monitoring Services to delegate the responsibilities of monitoring and releasing patches, bug-fixes, and enhancements to external vendors	1	2	-	1
PC12. monitor and manage deployed application	3	6	-	1
PC13. continuously look for and resolve bugs in software application	3	7	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. plan for application updates and patches	1	2	-	1
PC15. plan for application upgrades	1	2	-	1
PC16. plan for the retirement of application and migration to a new application	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8219
NOS Name	Develop software applications and manage application lifecycle for IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	5.5
Credits	3
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8245: Develop tools, processes and mechanisms for continuous integration and delivery for IoT solutions

Description

This unit is about developing tools, processes and mechanisms to assist continuous delivery and integration of developed solutions.

Scope

The scope covers the following :

- deploy production systems, monitor performance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Map IoT device types communication protocols and deployment scenarios to identify integration requirements
- PC2.** Identify hardware and software dependencies for IoT development and deployment
- PC3.** Define goals for CI/CD such as rapid updates secure deployments and scalability
- PC4.** Evaluate and select CI CD tools compatible with IoT including version control and containerization options
- PC5.** Integrate IoT-specific testing tools like simulators and hardware-in-the-loop testing into CI pipelines
- PC6.** Develop scripts to automate code compilation firmware image generation and signing
- PC7.** Incorporate static code analysis security checks and security scanning into build and deployment processes
- PC8.** Create automated test scripts for unit integration and system testing including hardware-in-the-loop tests
- PC9.** Automate firmware software deployment and use OTA mechanisms for deployment to real devices
- PC10.** Implement phased rollouts with rollback capabilities and integrate monitoring tools for device health and logs
- PC11.** Set up alerts for failures and anomalies and develop feedback loops to improve testing and code quality
- PC12.** Automate configuration management for different environments and maintain consistent versioning of firmware and software
- PC13.** Document CI CD workflows scripts and configurations along with SOPs for failures and updates
- PC14.** Run pilot pipelines on small IoT projects refine processes and plan gradual scaling
- PC15.** Collect data analyze results and automate additional testing for broader IoT deployments

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to DevOps
- KU2.** organizational policies and procedures for sharing data
- KU3.** who to involve when managing the administration of production systems and operations
- KU4.** different continuous delivery and deployment models
- KU5.** how to design and develop a staging environment
- KU6.** infrastructure as code processes
- KU7.** different configuration management tools such as Puppet, Chef
- KU8.** different continuous integration tools such as Jenkins, Travis CI, Bamboo
- KU9.** different application release automation tools such as BuildMaster, Puppet, FlexDeploy
- KU10.** different build automation tools such as Gradle, Jenkins
- KU11.** different application lifecycle management tools such as GitLab, JIRA
- KU12.** different methodological approaches to testing such as unit and integration testing
- KU13.** different test and QA automation tools such as selenium
- KU14.** how to work on various operating systems such as Linux, Ubuntu, Windows or MacOSX

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately
- GS3.** follow rule-based decision-making processes
- GS4.** plan and organize the work to achieve targets and meet deadlines
- GS5.** refer anomalies to the supervisor
- GS6.** check the work is complete and free from errors
- GS7.** work independently and collaboratively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Map IoT device types communication protocols and deployment scenarios to identify integration requirements	2	3	-	1
PC2. Identify hardware and software dependencies for IoT development and deployment	2	3	-	1
PC3. Define goals for CI/CD such as rapid updates secure deployments and scalability	2	3	-	1
PC4. Evaluate and select CI CD tools compatible with IoT including version control and containerization options	2	3	-	1
PC5. Integrate IoT-specific testing tools like simulators and hardware-in-the-loop testing into CI pipelines	2	3	-	1
PC6. Develop scripts to automate code compilation firmware image generation and signing	2	3	-	1
PC7. Incorporate static code analysis security checks and security scanning into build and deployment processes	2	3	-	1
PC8. Create automated test scripts for unit integration and system testing including hardware-in-the-loop tests	2	3	-	1
PC9. Automate firmware software deployment and use OTA mechanisms for deployment to real devices	2	3	-	1
PC10. Implement phased rollouts with rollback capabilities and integrate monitoring tools for device health and logs	2	3	-	1
PC11. Set up alerts for failures and anomalies and develop feedback loops to improve testing and code quality	2	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Automate configuration management for different environments and maintain consistent versioning of firmware and software	2	4	-	2
PC13. Document CI CD workflows scripts and configurations along with SOPs for failures and updates	2	4	-	2
PC14. Run pilot pipelines on small IoT projects refine processes and plan gradual scaling	2	4	-	2
PC15. Collect data analyze results and automate additional testing for broader IoT deployments	2	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8245
NOS Name	Develop tools, processes and mechanisms for continuous integration and delivery for IoT solutions
Sector	IT-ITeS
Sub-Sector	
Occupation	Internet of Things
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8224: Develop simulations and manage tests for end-to-end IoT solutions

Description

This unit is about simulating the IoT solution prior to the development process and performing various tests across the various layers of the solution.

Scope

The scope covers the following :

- Manage simulations
- Perform device testing
- Perform gateway testing
- Perform platform testing
- Perform integration testing
- Perform end-to-end testing
- Carry out test-case automation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Simulate the IoT solution during development to gain early feedback and support rapid prototyping and scaling
- PC2.** Use simulations to manage end to end testing and ensure interoperability of IoT components and communications
- PC3.** Test device discovery communication processes and performance metrics such as battery life and signal strength
- PC4.** Perform functional and interoperability testing of device firmware and hardware
- PC5.** Test communication between devices and cloud services including multi-device and multi-protocol compliance
- PC6.** Conduct remote diagnostics performance and security testing of the entire IoT solution
- PC7.** Test IoT platform functionalities including performance load testing and end to end solution validation for standards compliance
- PC8.** Verify interactions within modules and interfaces of the IoT solution to ensure proper integration and behavior
- PC9.** Develop automation suites and infrastructure to execute UI API and device level tests efficiently
- PC10.** Ensure comprehensive testing through automation and perform functional and user experience assessments of the complete IoT solution

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** organizational policies, procedures, and guidelines that relate to developing and deploying software applications
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting test results
- KU4.** who to involve while performing tests and simulations
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** various sources of information available for designing tests and how to access them
- KU7.** difference between various types of testing such as unit, sub-system, system, automated, functional, non-functional
- KU8.** issues that may occur with the testing requirements and how to address these
- KU9.** principles of designing tests
- KU10.** different simulation tool providers such as Autodesk and how to use their tools
- KU11.** how to evaluate and select the best simulation tool for your purposes
- KU12.** different test automation tools such as Selenium and how to use them
- KU13.** how to create suitable tests for particular applications
- KU14.** how to check whether test cases are suitable to be automated
- KU15.** how to access, create and modify different types of: a. high-level scenarios b. test cases c. automatic scripts d. test data e. test plans
- KU16.** different simulation methodologies and procedures used to check the tests are working and are fit for purpose
- KU17.** how to analyze and use feedback to improve your simulations
- KU18.** scripting/programming of languages to follow test cases
- KU19.** the current practice followed in the design of tests for software and hardware systems
- KU20.** how to identify and refer anomalies
- KU21.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** ask for clarification and advice from appropriate people
- GS3.** listen effectively and communicate information accurately
- GS4.** follow rule-based decision-making processes
- GS5.** make decisions on suitable courses
- GS6.** plan and organize the work to achieve targets and meet deadlines
- GS7.** refer anomalies to the supervisor
- GS8.** apply problem-solving approaches to different situations
- GS9.** analyze the business impact and disseminate relevant information to others
- GS10.** apply balanced judgments to different situations
- GS11.** check the work is complete and free from errors
- GS12.** work independently and collaboratively

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Simulate the IoT solution during development to gain early feedback and support rapid prototyping and scaling	3	5	-	2
PC2. Use simulations to manage end to end testing and ensure interoperability of IoT components and communications	3	5	-	2
PC3. Test device discovery communication processes and performance metrics such as battery life and signal strength	3	5	-	2
PC4. Perform functional and interoperability testing of device firmware and hardware	3	5	-	2
PC5. Test communication between devices and cloud services including multi-device and multi-protocol compliance	3	5	-	2
PC6. Conduct remote diagnostics performance and security testing of the entire IoT solution	3	5	-	2
PC7. Test IoT platform functionalities including performance load testing and end to end solution validation for standards compliance	3	5	-	2
PC8. Verify interactions within modules and interfaces of the IoT solution to ensure proper integration and behavior	3	5	-	2
PC9. Develop automation suites and infrastructure to execute UI API and device level tests efficiently	3	5	-	2
PC10. Ensure comprehensive testing through automation and perform functional and user experience assessments of the complete IoT solution	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8224
NOS Name	Develop simulations and manage tests for end-to-end IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	5.5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8244: Understand functional & non-functional testing of IoT solutions

Description

This unit is about understanding the functional & non-functional tests for the solution

Scope

The scope covers the following :

- Create test cases, Run test cases, Perform unit testing, Perform system testing, Perform acceptance testing, Carry out performance testing, Perform usability testing

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Create or modify test scenarios and plans to cover all functional and non functional requirements of the IoT solution
- PC2.** Identify and develop automated test cases and scripts relevant to functional performance and security testing
- PC3.** Access or generate test data aligned with test scenarios and ensure data quality for comprehensive testing
- PC4.** Execute test cases including functional load stress endurance spike scalability and chaos testing and evaluate outcomes
- PC5.** Validate test plans scripts and results with relevant stakeholders to ensure thorough coverage and accuracy
- PC6.** Verify that individual module codes and the integrated IoT solution function as per specifications and business criteria
- PC7.** Gather user feedback and perform usability testing to improve the end user experience
- PC8.** Conduct performance tests such as load stress endurance spike scalability and chaos testing to assess robustness and resilience
- PC9.** Ensure compliance of the IoT solution with all defined standards and business requirements through testing results
- PC10.** Document testing processes outcomes and recommendations for continuous improvement and stakeholder communication

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to developing and deploying software applications
- KU2.** organizational policies and procedures for sharing data
- KU3.** organizational policies and procedures for documenting test results

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- KU4.** who to involve while performing tests and simulations
- KU5.** the range of standard templates and tools available and how to use them
- KU6.** various sources of information available for designing tests and how to access them
- KU7.** difference between various types of testing such as unit, sub-system, system, automated, functional, non-functional
- KU8.** issues that may occur with the testing requirements and how to address these
- KU9.** principles of designing tests
- KU10.** different simulation tool providers such as Autodesk and how to use their tools
- KU11.** how to evaluate and select the best simulation tool for your purposes
- KU12.** different test automation tools such as Selenium and how to use them
- KU13.** how to create suitable tests for particular applications
- KU14.** how to check whether test cases are suitable to be automated
- KU15.** how to access, create and modify different types of: a. high-level scenarios b. test cases c. automatic scripts d. test data e. test plans
- KU16.** different simulation methodologies and procedures used to check the tests are working and are fit for purpose
- KU17.** how to analyze and use feedback to improve your simulations
- KU18.** scripting/programming of languages to follow test cases
- KU19.** the current practice followed in the design of tests for software and hardware systems
- KU20.** how to identify and refer anomalies
- KU21.** how to work on various operating systems such as Linux, Ubuntu, or Windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** ask for clarification and advice from appropriate people
- GS3.** listen effectively and communicate information accurately
- GS4.** follow rule-based decision-making processes
- GS5.** make decisions on suitable courses
- GS6.** plan and organize the work to achieve targets and meet deadlines
- GS7.** refer anomalies to the supervisor
- GS8.** apply problem-solving approaches to different situations
- GS9.** analyze the business impact and disseminate relevant information to others
- GS10.** apply balanced judgments to different situations
- GS11.** check the work is complete and free from errors
- GS12.** work independently and collaboratively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Create or modify test scenarios and plans to cover all functional and non functional requirements of the IoT solution	3	5	-	2
PC2. Identify and develop automated test cases and scripts relevant to functional performance and security testing	3	5	-	2
PC3. Access or generate test data aligned with test scenarios and ensure data quality for comprehensive testing	3	5	-	2
PC4. Execute test cases including functional load stress endurance spike scalability and chaos testing and evaluate outcomes	3	5	-	2
PC5. Validate test plans scripts and results with relevant stakeholders to ensure thorough coverage and accuracy	3	5	-	2
PC6. Verify that individual module codes and the integrated IoT solution function as per specifications and business criteria	3	5	-	2
PC7. Gather user feedback and perform usability testing to improve the end user experience	3	5	-	2
PC8. Conduct performance tests such as load stress endurance spike scalability and chaos testing to assess robustness and resilience	3	5	-	2
PC9. Ensure compliance of the IoT solution with all defined standards and business requirements through testing results	3	5	-	2
PC10. Document testing processes outcomes and recommendations for continuous improvement and stakeholder communication	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8244
NOS Name	Understand functional & non-functional testing of IoT solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8238: Create technical documents and manuals

Description

This unit is about creating various types of technical documents and manuals including online configuration manuals, user helps, solution information, installation guides, etc.

Scope

The scope covers the following :

- Plan the documentation process
- Develop drafts for the document
- Publish the document

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the purpose and the scope of the activity for which technical documentation is to be produced
- PC2.** obtain information for the technical document from relevant sources and stakeholders
- PC3.** draft technical document ensuring that content is concise, complete, and easy to consume
- PC4.** review technical document content with relevant stakeholders and document owners
- PC5.** ensure that technical document is formatted and designed as per specifications
- PC6.** transfer technical documents to relevant stakeholders for sign-off and publishing
- PC7.** continuously review and update technical document

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines that relate to drafting and publishing technical documentation
- KU2.** organizational policies and procedures for sharing data
- KU3.** the range of standard templates and tools available and how to use them
- KU4.** the solution for which documentation is to be created
- KU5.** how to create clear and concise content
- KU6.** how to format and align document

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow instructions, guidelines, procedures, rules, and service level agreements
- GS2.** listen effectively and communicate information accurately

Qualification Pack

- GS3.** follow rule-based decision-making processes
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and meet deadlines
- GS6.** refer anomalies to the supervisor
- GS7.** apply problem-solving approaches to different situations
- GS8.** analyze the business impact and disseminate relevant information to others
- GS9.** apply balanced judgments to different situations
- GS10.** check the work is complete and free from errors

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Identify the purpose and the scope of the activity for which technical documentation is to be produced	5	11	-	3
PC2. obtain information for the technical document from relevant sources and stakeholders	5	8	-	3
PC3. draft technical document ensuring that content is concise, complete, and easy to consume	5	8	-	3
PC4. review technical document content with relevant stakeholders and document owners	3	4	-	2
PC5. ensure that technical document is formatted and designed as per specifications	3	4	-	3
PC6. transfer technical documents to relevant stakeholders for sign-off and publishing	3	4	-	3
PC7. continuously review and update technical document	6	11	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8238
NOS Name	Create technical documents and manuals
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Internet of Things
NSQF Level	5.5
Credits	1
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8218.Design, develop and maintain software across appropriate layers of IoT solutions	30	50	0	20	100	15
SSC/N8219.Develop software applications and manage application lifecycle for IoT solutions	30	50	0	20	100	15
SSC/N8245.Develop tools, processes and mechanisms for continuous integration and delivery for IoT solutions	30	50	0	20	100	15
SSC/N8224.Develop simulations and manage tests for end-to-end IoT solutions	30	50	0	20	100	15
SSC/N8244.Understand functional & non-functional testing of IoT solutions	30	50	0	20	100	15
SSC/N8238.Create technical documents and manuals	30	50	0	20	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	200	330	-	120	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.